

Tanvir Ahmed Fahim

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Summary

Electrical & Electronic Engineering graduate with strong technical, analytical, and problem-solving skills. Eager to apply my engineering knowledge and technical skills in a professional environment, contribute to organizational success, and build a strong career through continuous learning and development.

Education

2022 – 2026
Cumilla, Bangladesh

Bachelor of Science in Electrical & Electronic Engineering,
Bangladesh Army International University of Science & Technology (BAIUST)
CGPA: 3.81/4.00 (Thesis defense pending)

Professional Experience

02/09/2026 – 24/09/2026
Dhaka, Bangladesh

Industrial Trainee, *Nagorik TV*
- Gained practical exposure to television broadcasting operations, including studio production, control-room workflow, content management, and transmission systems.
- Developed an understanding of the end-to-end broadcast workflow involving Studio, PCR, MCR, Ingest, Editing, Newsroom, CER, Archive, and RF transmission

10/2020 – 05/2021
Cox's Bazar, Bangladesh

Wireless Operator, *Field Intelligence Unit*
- Operated secure wireless communication systems to support field intelligence and operational activities.
- Maintained reliable communication between teams while following communication protocols, security procedures, and operational instructions.

07/2019 – 04/2022
Bangladesh

Soldier, *Bangladesh Army*
- Served in signals units to support military communication networks and tactical field operations, strengthening discipline and teamwork across multiple operational roles.
- Developed the ability to work effectively under pressure while maintaining professionalism and operational readiness.

Skills

Technical Skills: Microsoft 365, CST Studio Suite, HFSS, COMSOL, ADS, LTSpice, PSpice, Proteus, PSCAD, MATLAB, Python, C, Arduino, AutoCAD, SolidWorks, Simulink, Origin, SigmaPlot, Visio.

Professional Skills: Teamwork & Collaboration, Leadership, Communication, Problem-Solving, Time Management, Adaptability, Presentation Skills, Technical Writing, Ability to Work Under Pressure.

Projects

07/2026

ESP32-Based Multimodal Health Monitoring System
- Designed and developed an ESP32-based multimodal health monitoring system integrating a MAX30100 pulse-oximeter, three-electrode EEG sensor, and OLED display for real-time physiological-signal acquisition, with Wi-Fi-enabled remote monitoring through the Blynk IoT platform

01/2025

Advanced Pipe Inspection Robot
- Designed an advanced pipe inspection robot for complex pipeline infrastructures, featuring a wall-press mechanism, a sensor suite, an adaptive control system, and real-time data transmission, offering a robust and scalable solution for modern infrastructure inspection.

12/2024

Smart Car Parking System
- Developed an IoT-based smart parking system utilizing IR obstacle sensors, servo motors, and LED indicators for automated gate control and real-time occupancy monitoring, reducing manual effort and improving efficiency.

08/2024

Design and Implementation of FIR Filters for Image Processing Applications
- Implemented FIR filter design techniques to enhance edge detection and reduce noise in image processing applications, improving accuracy in medical imaging diagnostics and security surveillance systems.

Languages

Bangla
Native language; excellent proficiency in speaking, reading, and writing.

English
Proficient in speaking, reading, writing, and professional communication.

Community Service & Volunteering Activities

11/2024 – 10/2025 **IEEE BAIUST Student Branch,** *Senior Executive Member*
- Contributed to student branch activities and supported the planning and coordination of academic and professional initiatives.
- Collaborated with fellow members to promote student engagement and organizational activities.

09/2024 – 10/2025 **BAIUST Robotics Society,** *General Secretary*
- Led and coordinated robotics society activities, technical events, and student engagement initiatives.
- Worked with team members to organize programs, manage responsibilities, and promote robotics and technology-based learning.

2017 – 2018 **Bangladesh National Cadet Corps (BNCC), FGC Platoon,** *Cadet*
- Participated in cadet training and organizational activities, developing discipline, leadership, teamwork, and a strong sense of responsibility.

2014 – 2016 **Shaheen Academy School and College Scout Group,** *Scout Member*
- Participated in scouting activities, parade drills, community service, and team-based programs, developing discipline, leadership, teamwork, and social responsibility.

Achievements & Honors

08/12/2025 **Best Poster Award,** *BAIUST Inter-Department Poster Presentation Competition*

2022 – 2024 **VC List Merit Tuition Waiver (10%–40%),** *BAIUST*
Awarded for academic excellence in the 1st–4th and 6th–7th semesters.

02/2016 **Scholarship on Board of Secondary School Certificate**

References

Dr. Mohammad Shafiul Alam, *Associate Professor & Head,*
Department of EEE, Bangladesh Army International University of Science and Technology
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Md. Abdullah-Al-Mamun, *Assistant Professor,*
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